

D618

Environmental Health

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A town of 12,000 draws water from a lake popular with summer boaters. Its conventional plant adds a coagulant, lets floc settle, and then uses chlorine. An engineer notes one standard barrier is missing between settling and chlorine. Which step is it?

- A. Coagulation.
- B. None; chlorine alone always makes water safe.
- C. A second dose of chlorine.
- D. Filtration.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

Monitoring along a river that supports a busy fishing charter business found a persistent pesticide at 0.00005 ppm in water, 0.04 ppm in plankton, 0.5 ppm in small fish, and 25 ppm in fish-eating birds. The survey was funded by a state wildlife grant. By what factor did the concentration increase from small fish to birds?

- A. 500,000
- B. 50
- C. 0.02
- D. 24.5

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A chemical is at 0.2 ppm in zooplankton and increases tenfold at each of three steps up the food chain. What is the concentration in the top predator?

- A. 0.0002 ppm
- B. 6 ppm
- C. 200 ppm
- D. 20 ppm

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A volunteer made a 20-quart pot of chili for a potluck of 150 guests at a community hall that was recently repainted. After the event, about 12 quarts are left, still hot. What is the safest way to store it?

- A. Divide it into shallow containers and refrigerate promptly.
- B. Put the full pot in the refrigerator with the lid on.
- C. It is safe for any length of time because it was cooked thoroughly.
- D. Leave the pot on the counter overnight to cool before refrigerating.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

Cooked rice sat out at a 75°F picnic for 3 hours. A guest suggests reheating it to 165°F. What is the right call?

- A. Refrigerate it now for tomorrow.
- B. Keep it; the 1-hour limit only applies below 90°F.
- C. Reheat to 165°F; that makes it safe.
- D. Throw it out.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A 3-year-old has a confirmed blood lead level of 12 µg/dL. The family rents a 1940s house with peeling window paint and recently adopted a rescue dog. The city water system meets federal standards. What is the most important public health step?

- A. No action is needed below 45 µg/dL.
- B. Test only the tap water, since water is the main lead source.
- C. Do a home environmental investigation and fix lead paint and dust hazards.
- D. Test the dog for lead first.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

Tomorrow's forecast AQI for fine particles is 128. A middle school with 400 students in its newly renovated stadium has a track meet planned, and several runners have asthma. What guidance fits this AQI?

- A. Precautions are never needed for healthy children.
- B. Everyone should stay indoors all day.
- C. Sensitive groups, such as students with asthma, should reduce long or heavy outdoor exertion.
- D. No precautions; air quality is good below 150.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A radiography technician, who has worked at the plant for eight years, measures 400 $\mu\text{Sv/h}$ at 1 meter from a sealed source. The source was delivered Tuesday by a licensed courier. What dose rate should she expect at 4 meters?

- A. 25 $\mu\text{Sv/h}$
- B. 400 $\mu\text{Sv/h}$, because the source did not change.
- C. 100 $\mu\text{Sv/h}$
- D. 6,400 $\mu\text{Sv/h}$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A source reads 900 $\mu\text{Sv/h}$ at 1 meter. How far away must a worker stand for the rate to drop to 100 $\mu\text{Sv/h}$?

- A. 3 meters
- B. Distance does not matter if any shielding is present.
- C. 0.33 meters
- D. 9 meters

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

Three mosquito pools in a suburb test positive for West Nile virus. The county fair, which draws 50,000 visitors, is next month. Many yards have clogged gutters, old tires, and birdbaths. What should be the foundation of the response?

- A.** Cancel the county fair.
- B.** Remove standing water and treat breeding sites with larvicide, with continued trapping.
- C.** Spray adult-mosquito insecticide weekly across the whole county as the main plan.
- D.** Spraying will eliminate all risk of infection.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A city with a new electric bus fleet expects more heat waves. Many older adults live alone in top-floor apartments without air conditioning. Which planned action is climate adaptation?

- A.** Hand out electric fans, which fully protect people in any heat.
- B.** Expand the electric bus fleet to cut emissions.
- C.** Install solar panels on city buildings.
- D.** Open cooling centers and run phone check-ins for older adults during heat alerts.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

Which of these is climate mitigation rather than adaptation?

- A.** Issuing wildfire smoke alerts.
- B.** Capturing methane from a landfill.
- C.** Raising a road above the expected flood line.
- D.** Planting shade trees in hot neighborhoods only for cooling.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

~~A town of 12,000 draws water from a lake popular with summer boaters.~~ Its conventional plant adds a coagulant, lets floc settle, and then uses chlorine. An engineer notes one standard barrier is missing between settling and chlorine. Which step is it?

Target word: missing treatment step **Decides:** Filtration sits between sedimentation and disinfection.

A	Confusion set	Already listed as the first step.
B	Absolute language	Chlorine does not remove particles and is weak against <i>Cryptosporidium</i> .
C	Adjacent category	More disinfectant is not a physical barrier and does little against some protozoa.
D	Correct	Filtration removes particles and cysts that settling misses and that chlorine handles poorly.

2. Answer B Concept 2

~~Monitoring along a river that supports a busy fishing charter business~~ found a persistent pesticide at 0.00005 ppm in water, 0.04 ppm in plankton, 0.5 ppm in small fish, and 25 ppm in fish-eating birds. ~~The survey was funded by a state wildlife grant.~~ By what factor did the concentration increase from small fish to birds?

Target word: biomagnification factor, small fish to birds **Decides:** Factor = $25 / 0.5$.

A	Wrong stem number	Uses the water level ($25 / 0.00005$), not small fish.
B	Correct	$25 / 0.5 = 50$.
C	Confusion set	Flips the ratio ($0.5 / 25$), which would mean levels fall up the chain.
D	Inverted operation	Subtracts instead of dividing.

3. Answer C Concept 2 · transfer

A chemical is at 0.2 ppm in zooplankton and increases tenfold at each of three steps up the food chain. What is the concentration in the top predator?

Target word: solve for top-level concentration **Decides:** $0.2 \times 10 \times 10 \times 10$.

A	Inverted operation	Divides at each step, as if levels fell.
B	Adjacent category	Treats the increase as additive (0.2×30) rather than multiplying at each step.
C	Correct	$0.2 \times 1,000 = 200$ ppm.
D	Dropped element	Applies only two of the three steps.

4. Answer A Concept 3

~~A volunteer made a 20-quart pot of chili for a potluck of 150 guests at a community hall that was recently repainted.~~ After the event, about 12 quarts are left, still hot. What is the safest way to store it?

Target word: safe cooling **Decides:** A large, deep batch cools too slowly; it must be split to leave the danger zone quickly.

A	Correct	Shallow containers let food pass through the danger zone fast.
B	Dropped element	A large, deep, covered mass cools too slowly in the center.
C	Absolute language	Spores survive cooking and grow as food cools.
D	Inverted operation	Keeps the food in the danger zone for hours.

5. Answer D Concept 3 · transfer

Cooked rice sat out at a 75°F picnic for 3 hours. A guest suggests reheating it to 165°F. What is the right call?

Target word: time rule and reheating limits **Decides:** Over 2 hours in the danger zone; some bacterial toxins survive reheating.

A	Dropped element	Chilling stops further growth but does not undo 3 hours in the danger zone.
B	Inverted operation	The 1-hour limit applies above 90°F; the 2-hour limit applies here.
C	Adjacent category	Reheating kills many bacteria but not heat-stable toxins already formed.
D	Correct	It exceeded 2 hours, and heat-stable toxins can remain after reheating.

6. Answer C Concept 4

A 3-year-old has a confirmed blood lead level of 12 µg/dL. The family rents a 1940s house with peeling window paint ~~and recently adopted a rescue dog~~. The city water system meets federal standards. What is the most important public health step?

Target word: priority response **Decides:** Old peeling paint is the likely source; removing the source prevents further exposure.

A	Wrong stem number	45 is a common threshold for chelation; no level is safe and 12 is well above the reference value.
B	Confusion set	For young children in older housing, paint and dust are the usual main source.
C	Correct	Removing the source is the core of lead case management.
D	Camouflage	The pet detail does not change the priority of finding the home source.

7. Answer C Concept 5

Tomorrow's forecast AQI for fine particles is 128. A middle school ~~with 400 students in its newly renovated stadium~~ has a track meet planned, and several runners have asthma. What guidance fits this AQI?

Target word: AQI category action **Decides:** 128 is unhealthy for sensitive groups.

A	Absolute language	Children are themselves a sensitive group.
B	Adjacent category	That fits much higher AQI levels, not 101-150.
C	Correct	101-150 targets sensitive groups.
D	Inverted operation	Only 0-50 is good; above 100 affects sensitive groups.

8. Answer A Concept 6

A radiography technician, ~~who has worked at the plant for eight years~~, measures 400 µSv/h at 1 meter from a sealed source. ~~The source was delivered Tuesday by a licensed courier~~. What dose rate should she expect at 4 meters?

Target word: dose rate at 4 m **Decides:** Inverse square: 400 / 4².

A	Correct	400 / 16 = 25.
B	Confusion set	Source strength is the same, but dose rate depends on distance.
C	Dropped element	Divides by distance, not distance squared.
D	Inverted operation	Multiplies by the square instead of dividing.

9. Answer A Concept 6 · transfer

A source reads 900 $\mu\text{Sv/h}$ at 1 meter. How far away must a worker stand for the rate to drop to 100 $\mu\text{Sv/h}$?

Target word: solve for distance **Decides:** $900/100 = 9 = d^2$; $d = 3$.

A	Correct	Square root of 9 = 3 m.
B	Absolute language	Time, distance, and shielding all work together.
C	Inverted operation	Moving closer raises the rate.
D	Dropped element	Forgets the square root.

10. Answer B Concept 7

Three mosquito pools in a suburb test positive for West Nile virus. ~~The county fair, which draws 50,000 visitors, is next month.~~ Many yards have clogged gutters, old tires, and birdbaths. What should be the foundation of the response?

Target word: core vector strategy **Decides:** Breeding sites are visible; source reduction is the foundation of integrated management.

A	Camouflage	Reacts to the crowd size; does not address breeding sites.
B	Correct	Source reduction and surveillance come first in integrated management.
C	Adjacent category	Adulticiding is a targeted supplement, not the foundation.
D	Absolute language	No single measure removes all risk.

11. Answer D Concept 8

~~A city with a new electric bus fleet~~ expects more heat waves. Many older adults live alone in top-floor apartments without air conditioning. Which planned action is climate adaptation?

Target word: classify as adaptation **Decides:** Adaptation reduces harm from heat already occurring.

A	Absolute language	Fans are not protective at very high temperatures.
B	Camouflage	Mitigation, prompted by the bus detail; it does not protect people now.
C	Confusion set	Cuts emissions, which is mitigation.
D	Correct	Protects vulnerable people from existing heat risk.

12. Answer B Concept 8 · transfer

Which of these is climate mitigation rather than adaptation?

Target word: classify as mitigation **Decides:** Mitigation lowers greenhouse gas emissions.

A	Adjacent category	Adaptation; it reduces harm from smoke.
B	Correct	Cuts a potent greenhouse gas.
C	Inverted operation	Adaptation to flooding.
D	Confusion set	Framed for cooling, so adaptation, even though trees also store some carbon.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	6	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Multiple-barrier drinking water treatment	Conventional treatment runs coagulation and flocculation, sedimentation, filtration, then disinfection, so each barrier removes what the one before misses.	Thinking disinfection alone makes water safe, or that it comes first.	A question about the order of treatment steps or what a single step removes.
2	Bioaccumulation and biomagnification	Persistent, fat-soluble chemicals build up in organisms and multiply in concentration at each step up the food chain.	Adding the increases step by step instead of multiplying, or expecting the highest levels at the bottom of the chain.	Concentrations reported at several food-chain levels.
3	Time and temperature in food safety	Bacteria grow fastest in the danger zone (about 40–140°F); perishable food should not sit there more than 2 hours (1 hour above 90°F), and reheating does not destroy every toxin.	Believing reheating makes any food safe, or cooling large batches slowly on the counter.	Food left out for a stated time, or a large hot batch that needs cooling.
4	Childhood lead exposure	No blood lead level is known to be safe; for young children the main source is deteriorating lead paint and dust in older housing, so the priority is finding and removing the source.	Treating a level below a treatment threshold as safe, or assuming drinking water is always the main source.	A young child, an elevated blood lead result, and an older home.
5	Reading the Air Quality Index	Higher AQI means worse air: 0–50 good, 51–100 moderate, 101–150 unhealthy for sensitive groups, 151–200 unhealthy for everyone.	Telling everyone to stay inside at 101–150, or ignoring sensitive groups below 151.	An AQI number plus people with asthma, children, or older adults.
6	Time, distance, shielding (inverse square law)	Radiation intensity from a point source falls with the square of distance: double the distance, one-quarter the dose rate.	Assuming intensity falls in a straight line with distance.	A dose rate at one distance and a question about another distance.
7	Integrated vector management	Surveillance and source reduction (removing standing water, larviciding breeding sites) come first; adult-mosquito spraying is a targeted add-on when risk is high.	Relying on routine broad spraying as the main strategy.	Positive mosquito pools or a vector-borne case, and a menu of control options.
8	Climate adaptation vs. mitigation	Mitigation cuts greenhouse gas emissions; adaptation reduces harm from climate changes already happening, such as heat alerts and cooling centers.	Calling any climate-related action mitigation.	An action that either lowers emissions or protects people from heat, floods, or smoke.

Course scope (public description): Environmental Health provides learners with foundational knowledge, covering the relationship people have with their environment, the risk management choices that are made, and

the resulting associations that affect the health and physical well-being of individuals, communities, and susceptible populations. There are no prerequisites for this course.